Physical Properties

Physical Properties

Physical properties of water controlled by interaction between water molecules.

Density (ρ)

- 1.0 $g \cdot cm^{-3}$, 1000 $kg \cdot m^{-3}$, 62.4 $lbs \cdot ft^{-3}$
- be aware of different mass/force definitions for Imperial Units
- Water density varies over narrow range
  - temperature (max ρ at 4 °C)
  - salinity (seawater ρ is 1.025 $g \cdot cm^{-3}$)
  - pressure
- $$\rho = \frac{1 - T - 288.9414}{508929.2 \cdot (T + 68.12963)(T - 3.9863)^2}$$

$$\rho_s = \rho + A \cdot S + B \cdot S^{1.5} + C \cdot S^2$$

$$A = 8.24493 \cdot 10^{-1} - 4.0899 \cdot 10^{-3} \cdot T + 7.6438 \cdot 10^{-5} \cdot T^2 - 8.2467 \cdot 10^{-7} \cdot T^3 + 5.3675 \cdot 10^{-9} \cdot T^4$$
$$B = -5.724 \cdot 10^{-3} + 1.0227 \cdot 10^{-4} \cdot T - 1.6546 \cdot 10^{-6} \cdot T^2$$
$$C = 4.8314 \cdot 10^{-4}$$

McCutcheon, S.C., Martin, J.L., Barnwell, T.O. Jr. 1993. Water Quality in Maidment, D.R. (Editor). Handbood of Hydrology.

Density (ρ)

- Compressibility of water (β)
- density change with pressure
- varies with T
  - at 0 °C, $4.9 \cdot 10^{-10}\ m^2 \cdot N^{-1}$
  - at 20 °C, $4.5 \cdot 10^{-10}\ m^2 \cdot N^{-1}$

$$\beta dP = \frac{d\rho_{H_2O}}{\rho_{H_2O}}$$
$$\beta dP = -\frac{dV_{H_2O}}{V_{H_2O}}$$

Density

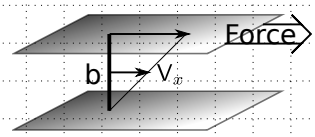
Question?
A one liter sample of water (10 °C) is collected from the bottom of a 500 ft deep well (well water level at ground surface).

- What is the density of water at this depth?
- What is the volume of the water sample at the ground surface?

Density

$$P = (500 \cdot ft) \cdot g \cdot 1000 \cdot (kg \cdot m^{-3})$$
$$P = 500000.0 \cdot kg \cdot ft \cdot g \cdot m^{-3}$$
$$P = 1494533.46 \cdot N \cdot m^{-2}$$
$$dV = (1 \cdot L) \cdot (4.7 \cdot 10^{-10} \cdot m^2 \cdot N^{-1}) \cdot P$$
$$dV = 0.000702 \cdot L$$

Viscosity



- Resistance to a shear stress (τ)
- μ is the shearing force per area (shear stress) that causes a unit change in velocity per unit thickness
- Dynamic (μ) vs. Kinematic Viscosity (ν)

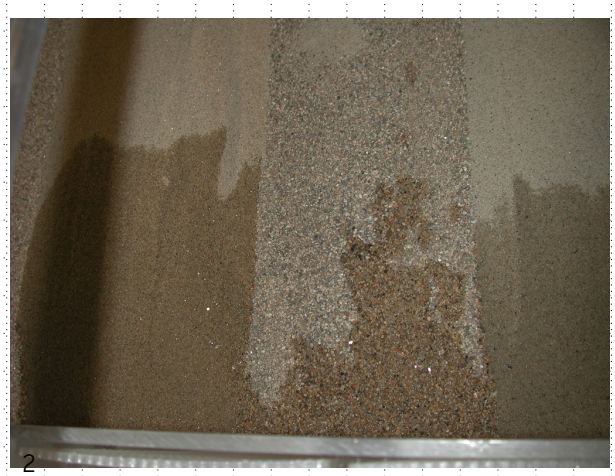
$$\frac{F}{A} = \mu \frac{dv_x}{db}$$
$$\nu = \frac{\mu}{\rho}$$

Viscosity

- Slows water movement through pore spaces
- Temperature dependent
  - at 0 °C, $1.79 \cdot 10^{-3}\ N \cdot sec \cdot m^2$
  - at 20 °C, $1.01 \cdot 10^{-3}\ N \cdot sec \cdot m^2$

Surface Tension

- internal cohesion
- 'wetting' on surfaces
- capillarity



Porosity

- fraction of void space in material (n)
- void ratio (e): ratio of voids and solid volumes
- primary vs. secondary porosity
- effective porosity

$$n = \frac{V_{void}}{V_{total}}$$

$$e = \frac{V_{void}}{V_{solid}}$$

Water Content

- Volumetric water content
- specific yield and specific retention
- dry and total bulk density

Hydraulic Head

Hydraulic Head

- Elevation of open water in equilibrium with atmosphere
- Measure of energy
- Components (Bernoulli Eq.)
  - Pressure
  - Gravity/Elevation
  - Velocity
- Head is energy normalized by weigh $(m \cdot g)$ of liquid

$$E = P \cdot V + m \cdot g \cdot z + \frac{m \cdot v^2}{2}$$

$$h = \frac{P}{\rho \cdot g} + z + \frac{v^2}{2 \cdot g}$$

Hydraulic Head

- Bernoulli's Equation assumes:
  - Incompressible fluid (constant density)
  - Nonviscous flow (energy not lost to friction)
  - non-turbulant flow

Hydraulic Head

Question?
The top of a well has an elevation of 100 m. above mean sea level. A pressure transducer is placed in the well at a depth of 20 m and measures a water pressure of 10 psi. What is the water level in the well?

Hydraulic Head

$$top_{well} = 100.0 \cdot m$$
$$depth = 20.0 \cdot m$$
$$P = 10.0 \cdot lb \cdot g \cdot in^{-2}$$
$$\rho = 1000.0 \cdot kg \cdot m^{-3}$$
$$H_p = \frac{P}{\rho \cdot g}$$
$$H_p = 0.01 \cdot lb \cdot m^3 \cdot kg^{-1} \cdot in^{-2}$$
$$H_p = 7.03m$$
$$H = top_{well} - depth + Hp$$
$$H = 87.03m$$

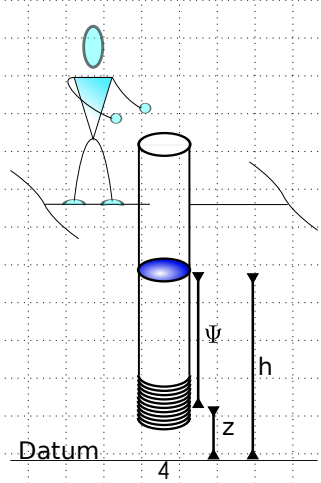
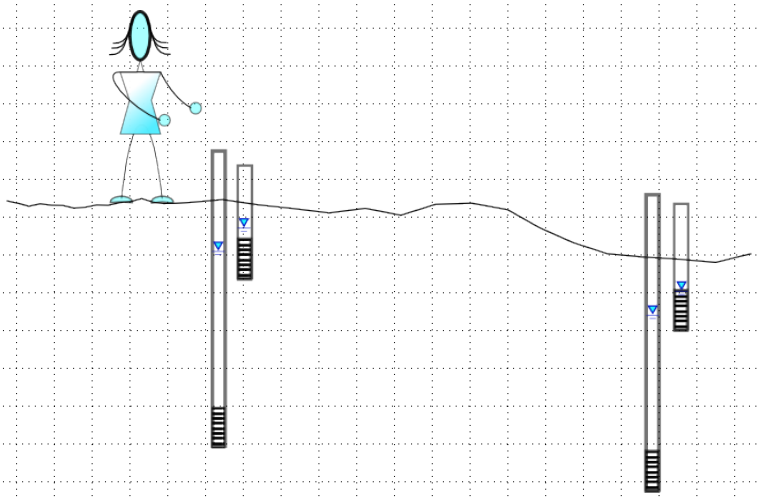
Hydraulic Head

- Velocity head is *usually* negligible in groundwater systems
- Water flows from high to low hydraulic head
- Wells measure h at opening

$$h = \Psi + z$$

×

Hydraulic Head



Hydraulic Head